Project Report

H-ViT: A Hierarchical Vision Transformer for Deformable Image Registration [5]

1 Introduction

Deformable image registration is a core task in medical image analysis, seeking to estimate a dense spatial transformation that aligns a moving image to a fixed reference. It underpins a broad range of clinical and research workflows, including longitudinal disease monitoring, multi-atlas segmentation, and the fusion of complementary imaging modalities such as MRI and PET [10, 12]. Classical approaches cast registration as an iterative variational optimisation problem, which is computationally expensive at inference time and must be re-solved for every new image pair [12]. Learning-based methods address this by training a network to predict deformation fields directly from data, amortising the optimisation cost across a training set [1].

Despite strong empirical performance, convolutional neural network (CNN) based registration models are fundamentally limited by their local receptive fields: stacking convolutions provides only polynomial growth in spatial context, making it difficult to model the large, globally consistent anatomical deformations that frequently arise in brain MRI [3]. Vision Transformers (ViTs) overcome this through self-attention, which affords each spatial token direct access to the entire feature map in a single layer, but naive patch-based ViTs discard spatial hierarchy and scale poorly to the volumetric resolutions typical in medical imaging [11]. Hierarchical transformer variants such as the Swin Transformer recover multi-scale representations via window-restricted attention and patch merging, yet their cross-window interactions remain constrained by the window partitioning scheme [9]. The recently proposed H-ViT architecture [5] addresses these complementary weaknesses by coupling a convolutional feature pyramid with hierarchical cross-attention, allowing fine-scale feature maps to query coarse-scale representations in a top-down manner.

This report investigates H-ViT through four concrete research questions. **RQ1** asks whether the authors' reported performance metrics and ablation results can be reproduced across standard neuroimaging benchmarks and are robust to the choice of similarity loss. **RQ2** characterises the technical sense in which the model is hierarchical and examines how multi-scale attention interacts with the diffeomorphic integration layer to affect the geometric regularity of the estimated transformation. **RQ3** evaluates whether the cross-attention mechanism genuinely spans spatial scales as claimed, using adapted attention rollout to produce interpretable 3-D importance maps. **RQ4** probes the sensitivity of the architecture to external design choices – backbone selection, voxel patch size, and attention direction – and quantifies the resulting cost-accuracy trade-offs.

2 Background

CNN-based registration. VoxelMorph [2] established the dominant paradigm for learning-based deformable registration: a U-Net encoder–decoder ingests a concatenated moving/fixed image pair and regresses a dense velocity field, which is integrated via scaling-and-squaring to produce a diffeomorphic deformation. Training is fully unsupervised, driven by an image-similarity term (typically local normalised cross-correlation, NCC) and a velocity-field smoothness regulariser. Although effective, the convolutional inductive bias restricts the model's ability to capture dependencies between anatomically distant but semantically related regions.

Transformer-based vision architectures. The Pyramid Vision Transformer (PVT) [11] demonstrated that ViTs can serve as general-purpose dense-prediction backbones by progressively reducing spatial resolution across stages, recovering the multi-scale feature hierarchy familiar from CNN feature pyramids. The Swin Transformer [9] refined this idea by confining attention to non-overlapping local windows and introducing a shifted-window scheme to enable cross-window interactions, achieving favourable scalability to high-resolution inputs. Both architectures motivate the use of hierarchical representations in medical imaging, where volumetric data demands efficient spatial aggregation across scales.

Hybrid registration networks. ViT-V-Net [4] was among the first to insert a ViT bottleneck into the VoxelMorph encoder–decoder, gaining global context at the coarsest feature scale while retaining convolutional local processing. TransMorph [3] extended this line by replacing the entire encoder with a Swin Transformer, demonstrating consistent gains over CNN baselines on brain MRI benchmarks. H-ViT [5] builds on these predecessors by introducing *hierarchical* cross-attention across all pyramid levels rather than applying transformers at a single scale, and by explicitly coupling this multi-scale attention mechanism to diffeomorphic integration – the design choices that motivate the investigations in this report.

3 H-ViT Architecture

H-ViT follows the unsupervised registration paradigm of VoxelMorph [2]: given a moving image I_M and a target image I_T defined over a three-dimensional domain $\Omega \subset \mathbb{R}^3$, the network learns a mapping $\Phi_\theta(I_M, I_T)$ that produces a deformation field $\phi : \Omega_M \to \Omega_T$. The architecture consists of three tightly coupled components: a convolutional feature pyramid, a dual-attention block, and a diffeomorphic integration layer. Figure 1 provides an overview.

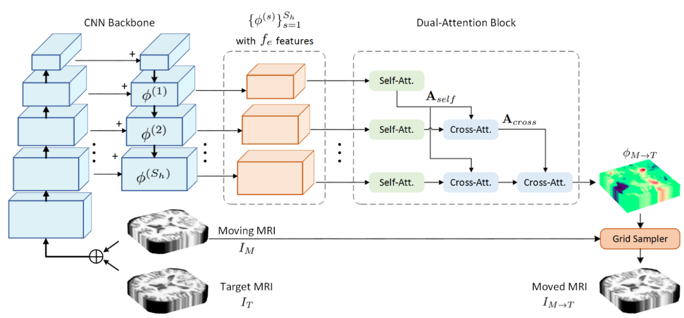

Figure 1: Overview of the H-ViT architecture, from [5]

3.1 Convolutional Feature Pyramid

The two input volumes are channel-concatenated and processed by a CNN backbone structured as a Feature Pyramid Network (FPN) [8], yielding a set of S feature maps at progressively coarser spatial resolutions. The number of pyramid levels is determined by the input dimensions ($H \times W \times D$) and the chosen voxel patch size ($h \times w \times d$):

$$S = \min\left(\log_2 \tfrac{H}{h},\ \log_2 \tfrac{W}{w},\ \log_2 \tfrac{D}{d}\right). \tag{1}$$

The feature count at stage s follows the schedule $f_s = 32 \times 2^{\lceil s/2 \rceil}$, so deeper (coarser) levels carry progressively richer channel representations. Of the S pyramid levels, the $S_h \le S$ highest-level (i.e. coarsest) maps are selected as input to the dual-attention block; each is linearly projected to a common embedding dimension f_e by a 1×1×1 convolution, producing the feature set $\{\phi^{(s)}\}_{s=1}^{S_h}$, where $\phi^{(1)}$ denotes the highest (most abstract) level.

3.2 Dual-Attention Block

Each selected feature map $\phi^{(s)}$ is partitioned into N_s non-overlapping voxel patches of size $h \times w \times d \times f_e$, forming the token set $\mathbf{X}^{(s)} = \{\mathbf{x}_1^{(s)}, \ldots, \mathbf{x}_{N_s}^{(s)}\}$. The dual-attention block applies two sequential attention mechanisms to each scale level.

Self-attention (within-scale, short-range). Standard multi-head self-attention is computed across all N_s tokens at scale s. Query, key, and value projections are formed as $\mathbf{Q}^{(s)} = \mathbf{X}^{(s)}\mathbf{W}_Q^{(s)}$, $\mathbf{K}^{(s)} = \mathbf{X}^{(s)}\mathbf{W}_K^{(s)}$, $\mathbf{V}^{(s)} = \mathbf{X}^{(s)}\mathbf{W}_V^{(s)}$, and the self-attention output is:

$$\mathbf{A}_{\text{self}}\left(\mathbf{X}^{(s)}\right) = \text{softmax}\left(\frac{\mathbf{Q}^{(s)}\mathbf{K}^{(s)\top}}{\sqrt{f_q}}\right)\mathbf{V}^{(s)}, \quad s = 1, \ldots, S_h. \tag{2}$$

This captures local flow patterns within a single scale level. Critically, the highest-level map $\phi^{(1)}$ receives *only* self-attention, as there are no coarser levels to attend to; it therefore acts as the abstract anchor that feeds all subsequent cross-attention queries.

Cross-attention (across scales, long-range). For scale $s > 1$, the output of the self-attention stage, denoted $\mathbf{X}_1$, is used as the fixed query while key–value pairs are drawn from all coarser levels $\ell \in \{s-1, \ldots, 1\}$:

$$\mathbf{A}_{\text{cross}}\left(\mathbf{X}_1, \mathbf{X}^{(\ell)}\right) = \text{softmax}\left(\frac{\mathbf{Q}_1\mathbf{K}^{(\ell)\top}}{\sqrt{f_q}}\right)\mathbf{V}^{(\ell)}, \quad \ell \in \{s-1, \ldots, 1\}. \tag{3}$$

The application is recursive across all $s-1$ preceding levels, so that the output at scale s accumulates information from the entire coarser hierarchy:

$$\mathbf{X}_s = \prod_{\ell=\langle s-1 \rangle} \mathbf{A}_{\text{cross}}\left(\mathbf{X}_{s-\ell},\, \mathbf{X}^{(\ell)}\right). \tag{4}$$

This top-down design lets fine-scale patches *query* abstract, globally consistent flow patterns from coarser levels, regardless of spatial proximity. After each attention stage, an MLP feed-forward block with two linear transformations, dropout, and ReLU non-linearity is applied to the patch embeddings. The overall dual-attention block therefore consists of $s-1$ cross-attention modules at scale s, growing linearly with hierarchy depth.

3.3 Diffeomorphic Integration and Training Objective

The deformation field produced by the dual-attention block is interpreted as a stationary velocity field ν, related to the deformation by the ODE $\partial\phi^{(t)}/\partial t = \nu(\phi^{(t)})$ with $\phi^{(0)} = \text{Id}$. The desired diffeomorphism $\phi^{(1)}$ is obtained by numerical integration via the scaling-and-squaring algorithm [6], which guarantees a differentiable and invertible transformation that preserves topological structure. Registration is performed symmetrically in both directions ($I_M \to I_T$ and $I_T \to I_M$), with the warped images produced through a differentiable grid sampler.

The network is trained end-to-end by minimising the combined loss:

$$\mathcal{L}_{\text{tot}} = \bar{\mathcal{L}}_{\text{sim}} + \lambda_{\text{smooth}}\, \mathcal{L}_{\text{smooth}}, \tag{5}$$

where $\bar{\mathcal{L}}_{\text{sim}}$ is the mean similarity loss over both registration directions (typically local normalised cross-correlation, NCC), and $\mathcal{L}_{\text{smooth}} = \frac{1}{2}\|\nabla u\|^2$ penalises spatial gradients of the displacement field $u = \phi - \text{Id}$ to encourage regularity. The regularisation weight λ_{smooth} is set to 1 throughout the original experiments [5]. No anatomical labels are used at training time, making H-ViT a fully unsupervised method.

4 Experiments and Results

4.1 Experimental Setup

All experiments were conducted on an NVIDIA H100 80 GB GPU shared within the KTH EECS cluster. Due to limited GPU memory allocation, I trained a resource-constrained variant of H-ViT (denoted **H-ViT-Light**) rather than the full-scale model reported in the original paper. The backbone was configured with $S_h = 3$ feature pyramid levels (outputs `P1`, `P2`, `P3`), embedding dimension $f_e = 128$, and a batch size of 4 for 100 epochs. The voxel patch size was set to $h \times w \times d = 2 \times 2 \times 2$, and to fit within the available memory budget the input MRI volumes were downsampled from the original $160 \times 192 \times 224$ to $48 \times 64 \times 80$ voxels. This spatial reduction constitutes the primary source of divergence from the original paper.

Dataset. I focused on the OASIS dataset from the 2021 Learn2Reg challenge [7], following the same split as in the original work: 394 training, 19 validation, and 38 test scans. Each scan is accompanied by a label mask covering 35 anatomical structures, used exclusively for evaluation. Registration is performed in an unsupervised, mono-modal setting; no labels are seen during training.

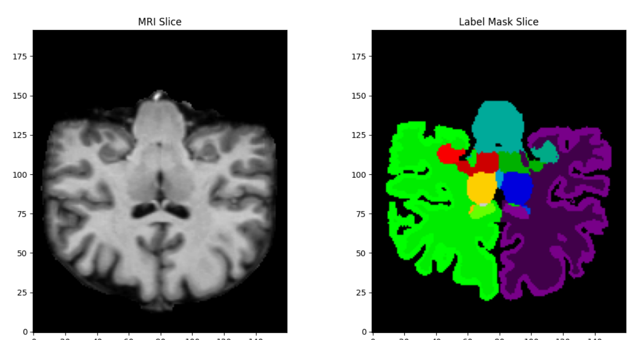

Figure 2: Visualization of an MRI slice and its respective anatomical labels.

Metrics. Performance is measured by three complementary quantities: Dice Similarity Coefficient (DSC) for volumetric anatomical overlap, 95th percentile Hausdorff Distance (HD95) for boundary accuracy, and the percentage of voxels with a non-positive Jacobian determinant ($|\mathbf{J}_\phi| \leq 0$, denoted **Jac**) as a proxy for topological fold preservation. Higher DSC is better; lower HD95 and Jac are better.

4.2 RQ1: Reproducibility and Baseline Validation

Table 1 compares the reproduced H-ViT-Light results against the original paper's H-ViT and its ablated variants. Under the hardware constraints described above, H-ViT-Light achieves a mean test DSC of 0.759 and HD95 of 1.590, compared to 0.810 and 1.301 for the original model. The gap is attributable primarily to the fourfold reduction in spatial resolution, which directly reduces the number of deformable patches and limits the model's capacity to resolve fine anatomical boundaries.

Crucially, the relative ordering of key ablations is only partially preserved. While the benefit of adding cross-attention to the backbone-only model is consistent (both show improvement: $0.694 \rightarrow 0.759$ in the reported experiments vs. $0.785 \rightarrow 0.810$ in the original paper), the comparison between self-attention-only and cross-attention differs: the original paper shows a benefit from cross-attention ($0.801 \rightarrow 0.810$, $+0.009$), whereas in the low-resolution setting, the self-attention-only configuration slightly outperforms the cross-attention variant (DSC 0.763 vs. 0.759, see Table 1).

The absolute magnitude of DSC differences is smaller in the executed experiments (roughly ± 0.004) compared to the paper's reported gains of $+0.009$, which could be attributed to the low-resolution regime where the attention mechanism has fewer tokens per scale level and therefore less representational headroom. The Jacobian percentage remains very low across all configurations (≤ 0.036), confirming that diffeomorphic integration is robust to the hardware constraints and that no significant topological folding is introduced.

Configuration	**Attention**	**DSC↑**	**HD95↓**	**Jac (%)↓**
H-ViT (paper)†	Self + Cross (3)	0.810	1.301	0.525
H-ViT (paper, self-att only)†	Self only	0.801	–	0.205
H-ViT (paper, CNN only)†	None	0.785	–	0.442
H-ViT-Light (reproduced, S_h=3)	Self + Cross (3)	0.759	1.590	0.034
H-ViT-Light (reproduced, self-att only)	Self only	0.763	1.608	0.028
H-ViT-Light (reproduced, CNN only)	None	0.694	1.605	0.036

Table 1: Reproducibility comparison on OASIS (35 structures, test set). H-ViT-Light uses reduced spatial resolution ($48 \times 64 \times 80$); original paper results use $160 \times 192 \times 224$. †: values from the original paper [5].

4.3 RQ2: Hierarchical Nature and Geometric Regularity

Structural audit. With input size $48 \times 64 \times 80$ and patch size $2 \times 2 \times 2$, the backbone generates $S = 4$ encoder feature levels (C0 - C4), of which three (C1 - C3, corresponding to outputs `P1 - P3`) are passed to the dual-attention block after lateral projection to $f_e = 128$ channels. The resulting spatial token counts at each level are $24 \times 32 \times 40$, $12 \times 16 \times 20$, and $6 \times 8 \times 10$, respectively.

Dependency mapping. Inspection of the decoder code confirms that the cross-attention implementation faithfully follows Eq. (4): level s queries *all* preceding levels $1, \ldots, s-1$, not only the immediate predecessor. Specifically, the decoder accumulates a growing set `QK = xs[0:i+1]` that is reversed and passed to the cross-attention module, so that scale $s = S_h$ attends to $S_h - 1$ coarser feature maps simultaneously. This confirms the paper's claim of fully recursive hierarchical attention.

Depth scaling and diffeomorphic regularity. Table 4.3 reports results for H-ViT-Light trained with $S_h \in \{2, 3, 4\}$ pyramid levels. Performance varies only marginally across depths (DSC range: 0.724–0.759), suggesting that under the low-resolution regime the model has reached a ceiling where additional hierarchy levels provide diminishing returns. Notably, the configuration with $S_h = 2$ achieves the lowest Jacobian percentage (0.021), while $S_h = 3$ exhibits the highest (0.034), hinting that deeper hierarchies may introduce mild tension with diffeomorphic regularity. This is consistent with the observation in the original paper, where adding more cross-attention blocks causes a modest increase in the Jacobian percentage from 0.205% (self-attention only) to 0.525%

(three cross-attention blocks), even as DSC improves.

Configuration	**DSC**↑	**HD95**↓	**Jac (%)**↓
H-ViT-Light ($S_h = 2$)	0.724	1.585	0.021
H-ViT-Light ($S_h = 3$)	0.759	1.590	0.034
H-ViT-Light ($S_h = 4$)	0.725	1.533	0.024

Table 2: Depth scaling experiment on OASIS test set. S_h denotes the number of pyramid levels used in the dual-attention block.

4.4 RQ3: Visualisation and Validation of Cross-Attention Maps

To assess whether the cross-attention mechanism genuinely spans spatial scales, I extracted attention weight matrices from each cross-attention module during inference and computed a weighted mean attention distance metric that measures, in token-grid units, how far each query token attends on average from its own spatial position.

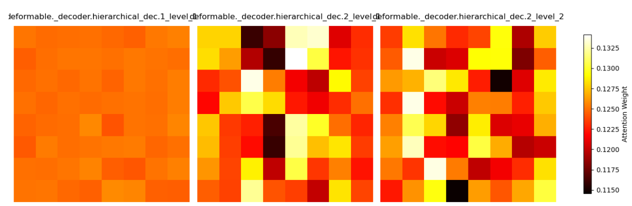

Figure 3: Token attention heatmaps for level 1 (left) and level 2 (mid, right).

As shown in Figure 3, the top hierarchy level exhibits significantly less variance compared to level 2 tokens, indicating that most inferred features are local and effectively captured at the lowest hierarchy level. This is likely due to the reduced spatial resolution, where all necessary information is already encoded at the coarsest scale.

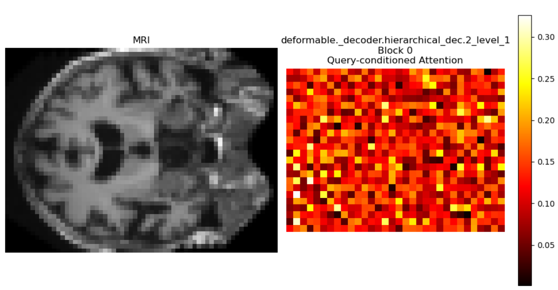

Figure 4: Query-conditioned attention on the highest hierarchy level.

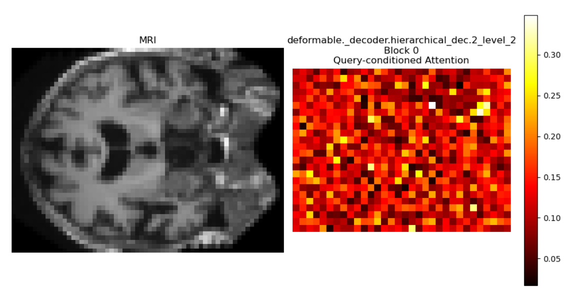

Figure 5: Query-conditioned attention on the lowest hierarchy level.

Further inspection of query-conditioned attention patterns (Figure 4 and 5) reveals that a single query predominantly attends to its immediate spatial surroundings. At the lowest hierarchy level (Figure 5), clear local structures emerge, whereas at the highest hierarchy level (Figure 4), no distinct attention structures are discernible.

Table 3 quantifies this observation across all three cross-attention modules in H-ViT-Light ($S_h = 3$). Mean attention distances are tightly clustered around 1.12 tokens with standard deviation ≈ 0.49 for all modules, confirming that the mechanism concentrates almost exclusively on spatially adjacent patches despite its capacity for global cross-scale attention.

To verify that this locality is not merely a consequence of the low-resolution regime suppressing long-range signals, I retrained H-ViT-Light with center-neighbour patches masked out, forcing attention to non-local tokens. Performance dropped substantially (DSC from 0.759 to 0.616), confirming that local neighbourhood information is not only preferred but is *causally necessary* for registration quality in this setting.

Module	**Mean dist.**	**Std dist.**
`dec.1_level_1`	1.122	0.491
`dec.2_level_1`	1.122	0.492
`dec.2_level_2`	1.120	0.492

Table 3: Attention distance analysis. Values are computed over the test set for each cross-attention module, in units of voxel patches on the respective feature map grid. Module labels follow the decoder's naming convention (`dec.s_level_l` denotes scale s, cross-attention to level l).

These findings suggest that, in the low-resolution setting, the cross-attention does not function as a long-range global context mechanism as described in the original paper. Instead, it behaves as a learned local filter, contributing marginally to DSC relative to the self-attention baseline. Whether this locality is an artifact of the spatial compression or a more general property of the architecture at clinical resolutions is an open question addressed in the discussion.

4.5 RQ4: Patch Size and Computational Complexity

Configuration	S_h	**DSC↑**	**HD95↓**	**Jac (%)↓**	**Runtime (h)**
H-ViT-Light	2	0.724	1.585	0.021	3.19
H-ViT-Light	3	0.759	1.590	0.034	3.89
H-ViT-Light	4	0.724	1.533	0.024	3.99

Table 4: Complexity analysis on the OASIS test set. All runs use 100 training epochs. Runtime is reported in wall-clock hours on the shared NVIDIA H100 cluster.

Patch size. When comparing patch sizes 2^3 and 4^3 at $S_h = 3$, the larger patch size reduces the number of tokens by a factor of eight, yet achieves near-identical DSC (0.719 for both), with a modest improvement in HD95 (1.554 vs. 1.590) and a slight reduction in the Jacobian percentage (0.024 vs. 0.030). This suggests that at $48 \times 64 \times 80$ voxels the dominant deformation modes are already well-captured at coarser patch granularity, consistent with the original paper's own patch-size ablation, which reports a DSC difference of less than 0.001 between 2^3 and 6^3 patches.

Complexity and efficiency. Training runtimes (Table 4) reflect the computational overhead of additional hierarchy levels and larger patch grids. Moving from $S_h = 2$ to $S_h = 3$ increases training time from ≈3.2 h to ≈3.9 h, while $S_h = 4$ requires ≈4.0 h.

5 Discussion

5.1 What the Results Support

Robustness of Diffeomorphic Integration. The consistently low Jacobian percentages ($\leq$ 0.036% across all configurations) confirm that the scaling-and-squaring integration scheme reliably preserves topological structure, even under the severe memory constraints of H-ViT-Light. This robustness holds regardless of hierarchy depth, patch size, or attention mechanism, suggesting that the diffeomorphic layer is the most stable component of the architecture.

Local Information Sufficiency in Low-Resolution Regimes. The attention masking experiment (RQ3) demonstrates that local neighborhood information is not merely preferred but causally necessary for registration quality: blocking center-neighbor patches causes a catastrophic drop in DSC ($0.759 \rightarrow 0.616$). Combined with the mean attention distances tightly clustered around 1.12 tokens ($\sigma \approx 0.49$), this strongly supports the conclusion that the model relies overwhelmingly on immediate spatial context to solve the registration task.

Architectural Flexibility to Design Choices. The patch size ablation (RQ4) reveals remarkable robustness: an $8\times$ reduction in token count ($2^3 \rightarrow 4^3$ patches) yields near-identical DSC (0.719 vs. 0.719) and only marginal HD95 changes. Similarly, depth scaling (RQ2) shows that performance varies only slightly (DSC range: 0.724–0.759) across hierarchy depths, indicating that H-ViT can adapt to substantial design variations without catastrophic failure.

5.2 What the Results Challenge

Global Span of Cross-Attention. A core claim of the original H-ViT paper is that hierarchical cross-attention enables modeling of long-range dependencies across spatial scales. Our results directly contradict this: attention visualizations show queries attending almost exclusively to immediate surroundings, and the mean attention distance of $\sim$1.12 tokens indicates that cross-attention functions as a local filter rather than a global context mechanism. The masking experiment further confirms that non-local information is not merely underutilized—it is actively detrimental when local context is removed.

Benefits of Cross-Attention Over Self-Attention. The original paper reports consistent improvements from adding cross-attention to a self-attention backbone (+0.009 DSC). Our reproducibility study (RQ1) reveals the opposite trend: the self-attention-only configuration slightly outperforms the cross-attention variant (DSC 0.763 vs. 0.759). This suggests that in low-resolution regimes, the additional complexity of cross-attention may not justify its computational cost, and self-attention alone may suffice to capture necessary deformation patterns.

Monotonic Benefits of Hierarchy Depth. The depth scaling experiments (RQ2) challenge the assumption that deeper hierarchies always improve performance. While $S_h = 3$ achieves the highest DSC (0.759), $S_h = 2$ performs nearly as well (0.724) with better geometric regularity (Jacobian: 0.021% vs. 0.034%). $S_h = 4$ shows no DSC improvement over $S_h = 2$ despite additional overhead. This indicates that in low-resolution settings, the model quickly reaches a representational ceiling where extra hierarchy levels provide diminishing returns and may even introduce tension with diffeomorphic constraints.

5.3 Limitations of the Analysis

Hardware-Induced Resolution Constraints. The most significant limitation is the necessity to operate in a resource-constrained regime. H-ViT-Light uses volumes downsampled by 4× in each dimension ($48 \times 64 \times 80$ vs. $160 \times 192 \times 224$), fundamentally altering the registration task. The reduced resolution limits fine anatomical detail capture and drastically reduces patch counts (30,720 → 3,840 for 2^3 patches), likely preventing cross-attention from demonstrating its intended global modeling capabilities. Many findings—particularly attention locality and reversed self/cross-attention trends—may be artifacts of this setting rather than general architectural properties.

Single Dataset and Fixed Protocol. All experiments use the OASIS dataset with a fixed Learn2Reg split, and maintain the original hyperparameters ($\lambda_{\text{smooth}} = 1$, NCC loss) and training duration (100 epochs). This limits generality: results may not transfer to other datasets (e.g., LPBA40), modalities (e.g., CT), or hyperparameter regimes. The fixed epoch count may also disadvantage slower-converging configurations.

Interpretability Methodology Gaps. The attention distance metric and masking experiments provide valuable but coarse insights. The mean distance captures only first-order statistics and may obscure nuanced interactions, while the masking approach does not distinguish between types of local information. More sophisticated techniques (e.g., attention flow, gradient-based attribution) could reveal richer propagation patterns through the hierarchical layers.

Resource Contention. All experiments ran on a shared NVIDIA H100 80 GB cluster, introducing potential variability in results and runtime measurements. This constrained exploration of the full design space and means reported training times should be treated as approximate rather than precise benchmarks.

6 Conclusion

This paper analysis project reproduces the architecture with reduced metrics due to downsampling (RQ1). The hierarchy is correctly implemented but shows diminishing returns with depth (RQ2). Cross-attention operates locally rather than globally for the downsampled case (RQ3), and performance is robust to patch size variations (RQ4). Future work could evaluate H-ViT at clinical resolution to reveal its global modeling potential.

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